



BRAC University
Department of Computer Science and Engineering

CSE 443: Bioinformatics-I (Set A)

Mid Exam: Fall 2025 Time: 1 hour 30 minutes Minutes Marks: 40

Name		ID		Section	
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1. (a) Explain alternative splicing and its implication in cells. [3]
(b) What is a clump? What role does it play in finding origin or replication? [3]
(c) Construct the skew table / diagram for this string: GCGGTAGCCGCGTAAGCCGGTACGC-CGT and indicate the possible location (s) of OriC. [4]
2. (a) Consider the following reads: CGC, CGT, CGT, CGT, GCG, GTC, GTC, TCG, TCG, TCG. Now construct overlap graph and de-bruijn graph from these reads. [3+3]
(b) What are the necessary conditions to find a euler path in a graph? [2]
(c) You have larger reads from an illumina genome sequencing experiments, each read is of 150bp. The quality is good. However, you dont have full coverage, means all the reads possible from the genome is not present. However, all the reads that you have got show good overlap. Is there any way to get a full coverage? How? [2]
3. (a) How does pegen hole principle helps in genome read mapping?
(b) Using the Boyer-Moore string-matching algorithm, illustrate how the pattern is searched within the text and quantify the savings compared to the naive algorithm. [6]
Text: ATGCGATATCGATGCGATC
Pattern: GCGAT
4. (a) Consider the following points (0,0), (0, 1), (-1, 2), (2, 0), (3, 0), (4, -1) for soft k -means clustering. Use the first and last data points as initial centers ($k = 2$). The formula for membership calculation and centroid update are given below.

membership	$w_{ik} = \frac{e^{-\beta \ x_i - \mu_k\ ^2}}{\sum_{j=1}^K e^{-\beta \ x_i - \mu_j\ ^2}}$	centroid update	$\mu_k = \frac{\sum_{i=1}^N w_{ik} x_i}{\sum_{i=1}^N w_{ik}}$
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perform the following considering $\beta = 0.8$:

- i. Find the membership of each of these data points to the clusters. [4]
ii. Update the cluster centers. [4]
- (b) There are 200 samples involved in a rna-seq experiment taken from human subjects. Among them 100 are healthy and the rest are infected with a new virus. Your experiment data is giving you gene expression count for 22000 genes for each. How do you find the genes that are most important in distinguishing these two group of patients? [2]